

Ming Zhong

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SUMMARY

- Senior Data Scientist with 5+ years in the retail industry, specializing in turning complex data into strategic insights.
- Proven track record of agile collaboration, building ETL pipelines, and delivering data-driven solutions.
- Active researcher in Agentic and Generative AI; co-author of peer-reviewed papers at top-tier conferences.

PROFESSIONAL EXPERIENCE

Senior Data Scientist, Circle K Stores Inc., Tempe, AZ

Aug. 2022 - Present

- **Analogous Site Tool:** Identified comparable stores for financial forecasting using Mahalanobis distance and ML-driven feature selection, enabling data-driven site analysis for Real Estate teams.
- **White Space Tools:** Developed random forest ROI models for 220K+ block groups using location-based features, identifying 15K+ high-ROI areas, which optimized site planning in US/Canada.
- **Site Selection Model:** Enhanced revenue prediction accuracy by 15% through the development of a random forest regression model, preventing potential real estate investment failures exceeding \$2M in the Texas sector.
- **Churn Prediction:** Built end-to-end churn prediction pipeline (Scikit-Learn, MLflow) with > 0.9 accuracy and 0.82 AUC. Analyzed customer segments and visualized key churn drivers to support retention strategies.
- **Carwash Demand Forecast:** Forecasted carwash demand using Prophet, ARIMA, RNNs, and LSTM models, enabling supply chain and pricing optimization that led to \$500K+ in cost savings.

Data Scientist, Circle K Stores Inc., Tempe, AZ

June 2021 - July 2022

- **Localized Pricing Engine:** Migrated pricing engine from Databricks to Azure Data Factory, reducing refresh time by 88% (from 80 to 10 hours) and operational cost by 76% (\$500 each run).
- **Localized Pricing:** Modeled SKU-level price elasticity using UCM-based time series forecasting, driving 3% margin lift and over \$120M in annual gross profit.
- **Cigarettes A/B Testing:** Designed and executed A/B testing framework to evaluate cigarette price changes in pilot stores, generating \$700K in margin uplift across the Midwest.
- **ETL Pipeline:** Engineered data pipelines to track the state of the storage and ensure data quality for modeling and business purposes, eliminating over 60 hours of manual labor per quarter.

Business Analyst, ROKT Corp., New York, NY

Jan. 2020 - May 2021

- **Conversions Analytics:** Analyzed attribution identifier performance via SQL on large-scale campaign data; recommended inclusion of phone numbers, increasing ad conversions by 40%.
- **COVID Dashboard:** Designed Tableau dashboards visualizing 10M+ transactions for 10+ clients to support ad relaunch decisions with COVID recovery metrics.
- **Account Managers ChatBot:** Built an NLP chatbot in Google Chat (Python, spaCy, NLTK) to automate client campaign insights; won 1st place in internal hackathon and explored for broader team use.

PUBLICATIONS

AssistedDS: Measuring How External Expert Knowledge Assists LLMs in Automated Data Science

Empirical Methods in Natural Language Processing (EMNLP) 2025, Under Review

EDUCATION

Georgia Institute of Technology

Master of Science, Computer Science

Atlanta, GA

Aug. 2023 - Present

Columbia University

Master of Arts, Statistics

New York, NY

Dec. 2019

University of Macau

Bachelor of Science, Finance

Macau, China

Jul. 2018

SKILLS

Languages/Tools: Python, SQL, R, Spark, Java, JavaScript, HTML, Tableau, PowerBI

Frameworks: Agile, Pyspark, TensorFlow, MLflow, Keras, Scikit-learn, Pandas, PyTorch, nloptr

Platforms: Databricks, AWS (certified solution architect), GCP, Azure, Git